

Plant model and ground truth: what it does and why it is correct

July 2026

Summary. The AROMA 5GDHC simulator is the thesis ground truth: 23 nodes, 29 edges, 5 prosumer substations, 736 states. Six 24-hour scenarios, a source-pulse transport-delay experiment, and a resolution sweep show it behaves like a real heating network on both sides of the capacity limit. All nine coded invariants pass on the shipped results; most clear their bound by orders of magnitude, and the two that do not (the flow-balance transient and the extracted-over-source ratio) are reported against their bound in Section 4.

1 What this component does and what I did

The plant is the AROMA 5GDHC network simulator in heating mode: 23 nodes, 29 edges, 5 prosumer substations, 736 continuous-time states (mostly the discretized pipe cells), integrated by `ode45`. The production sample step is $T_s = 900$ s (`shared/params.m`); the demos here read out at 60 s so the transients are visible. It is both plant and ground truth: every predictor is identified on its trajectories, every controller judged against it.

The pipe transport, the node mixing and the source model are the AROMA reference physics and I left them as they are. I extended the simulator where the formulation needs it, and those extensions are mine: the per-edge flow state (`shared/plant/flow_dynamics.m`, driven through the r_q path in `simulate_plant.m`), the substation return-temperature floor and the $c \leq d$ saturation (`prosumer_house.m`), a solver-tolerance override used only by the resolution sweep, and a correction to the advection speed in the reverse-flow branch of the upwind scheme (`pipe_rhs_bidir.m`), which the heating-mode runs never exercise. On top of that I built the validation layer: six 24-hour scenarios (`demo_plant.m`), nine coded invariants (`tests/validate_plant.m`), a source-pulse experiment (`demo_pulse.m`), and a resolution sweep (`verify_ode_resolution.m`). T1–T9 number the invariants; P1–P4 the four pulse checks.

Four pieces of physics carry the component and are what the predictor must learn.

Flow dynamics (flows are states). Each edge flow tracks its command first order,

$$q_{k+1}^{(e)} = a^{(e)} q_k^{(e)} + (1 - a^{(e)}) r_{q,k}^{(e)}, \quad a^{(e)} = e^{-T_s/\varepsilon^{(e)}},$$

with time constant $\varepsilon = 600$ s (supply/return edges), 300 s (user stubs); zero-order hold in `simulate_plant.m` line 178. Flows are genuine states, not instantaneous inputs.

Pipe transport. Each pipe solves advection-loss

$$\frac{\partial T}{\partial t} = -v \frac{\partial T}{\partial x} - \alpha (T - T_{\text{ext}}), \quad v = \frac{q}{\rho A_{cs}},$$

by first-order upwind on $dx = 10$ m cells with loss coefficient $\alpha = 10^{-6}$ 1/s. Heat moves at the bulk velocity.

Substation clip (the feasibility contract).

$$c^{(i)} = \min(d^{(i)}, c_p q^{(i)} (T^{(i,s)} - T_{\min}^{(r)})), \quad T^{(i,r)} = T^{(i,s)} - \frac{c^{(i)}}{c_p q^{(i)}},$$

with return floor $T_{\min}^{(r)} = 15^\circ\text{C}$ (`prosumer_house.m` lines 27–33). Delivery saturates at $c_{\max} = c_p q (T^{(s)} - T_{\min}^{(r)})$, the return never breaks the floor, and any shortfall is booked as unmet demand. This is the contract the controller must respect.

Node mixing and source. Junctions mix flow-weighted; the source is a CSTR with outlet mixing $T_{\text{int}} = (1 - \theta_1)T_{\text{in}} + \theta_1 T_{\text{out}}$, mixing weight $\theta_1 = 0.7$. This gives a steady command-to-outlet gain $k_Q / (c_p q + k_Q) / \theta_1 \approx 1.09$ (source conductance k_Q , nominal flow q), matching the ≈ 1.09 gain measured in the pulse experiment.

The six scenarios share one demand profile and a 90-minute warmup, chosen so both the comfortable and the capacity-binding regimes appear. Demand met is the daily delivered-over-demanded energy ratio over all five consumers (Table 1).

scenario	flow	source supply command	demand met [%]
nominal_midday	1.0×	20 °C	100.00
morning_ramp	1.0×	20 °C	100.00
low_flow	0.5×	20 °C	81.22
high_flow	1.5×	20 °C	100.00
supply_step	1.0×	20 → 24 → 18 °C, 8 h phases	90.06
variable_flow	0.5 → 1.5 → 0.5×	ramp 20 °C	81.48

Table 1: The six 24-hour scenarios and their demand-met percentages (from `run_all_log.txt`).

Code layout

file	role
<code>shared/plant/build_plant.m</code>	builds the 23-node, 29-edge network and initial state
<code>shared/plant/simulate_plant.m</code>	integrates one scenario (<code>ode45</code>), samples at T_s
<code>shared/plant/flow_dynamics.m</code>	the edge-flow map $q_{k+1} = a q_k + (1 - a) r_{q,k}$
<code>demo_plant.m</code> + <code>plot_plant_demo.m</code>	run the six scenarios, save <code>demo.mat</code> , render dashboards
<code>demo_pulse.m</code>	the source-pulse delay experiment, saves <code>pulse.mat</code>
<code>tests/validate_plant.m</code>	the nine coded invariants of Section 4

2 How to reproduce

Everything runs from the repository root in MATLAB R2024b or later (run on R2025b; see README).

- **run_all_tests:** re-checks the nine invariants against the shipped `demo.mat` without re-simulating (seconds), prints PASS per invariant and scenario, writes `run_all_log.txt`. The build prints 23 nodes / 29 edges / 736 states, so that number is code-produced.
- **demo_plant:** re-simulates the six scenarios (about half an hour), saves `demo.mat`, renders the dashboards plus `demo_invariants`.

- `demo_pulse` (about 2 min): 24-hour warmup then a 25-hour run with a +5 K one-hour source-command pulse, saves `pulse.mat` and `pulse.pdf`, runs its four checks inline.
- `visualizations/1_plant/verify_ode_resolution` runs live against the build and hard-asserts deviations below 10^{-6} °C.

3 The figures

3.1 `demo_low_flow.pdf`: the clip engages

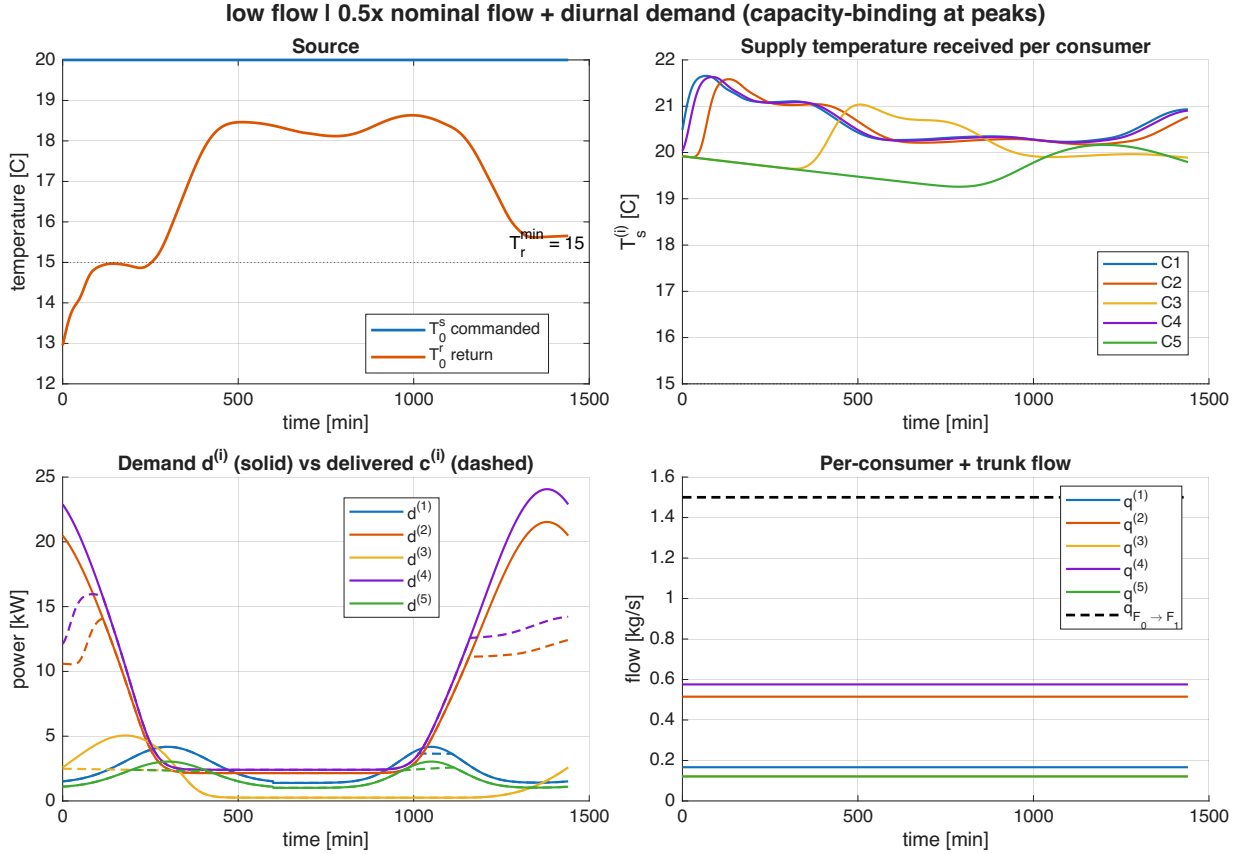


Figure 1: Low-flow scenario (0.5 $\times$): delivered heat is pinned below demand at the peaks while the return rides the 15 °C floor exactly, and recovers off-peak.

At half flow $c_{\max}$ halves, so at the peaks the substation delivers $c_{\max}$ instead of d , pins the return at 15 °C, and books the shortfall; off-peak it returns to parity. The aggregate return starts near 13 °C from the cold fill and 90-minute warmup (return pipes still flushing); the floor applies to the substation outlets, not this source-side mixture. The early received-temperature bumps are that flush reaching each tap with its own delay: C1/C2/C4 within two hours, C3 around minute 450, C5 late in the day.

3.2 demo_variable_flow.pdf: flows are states

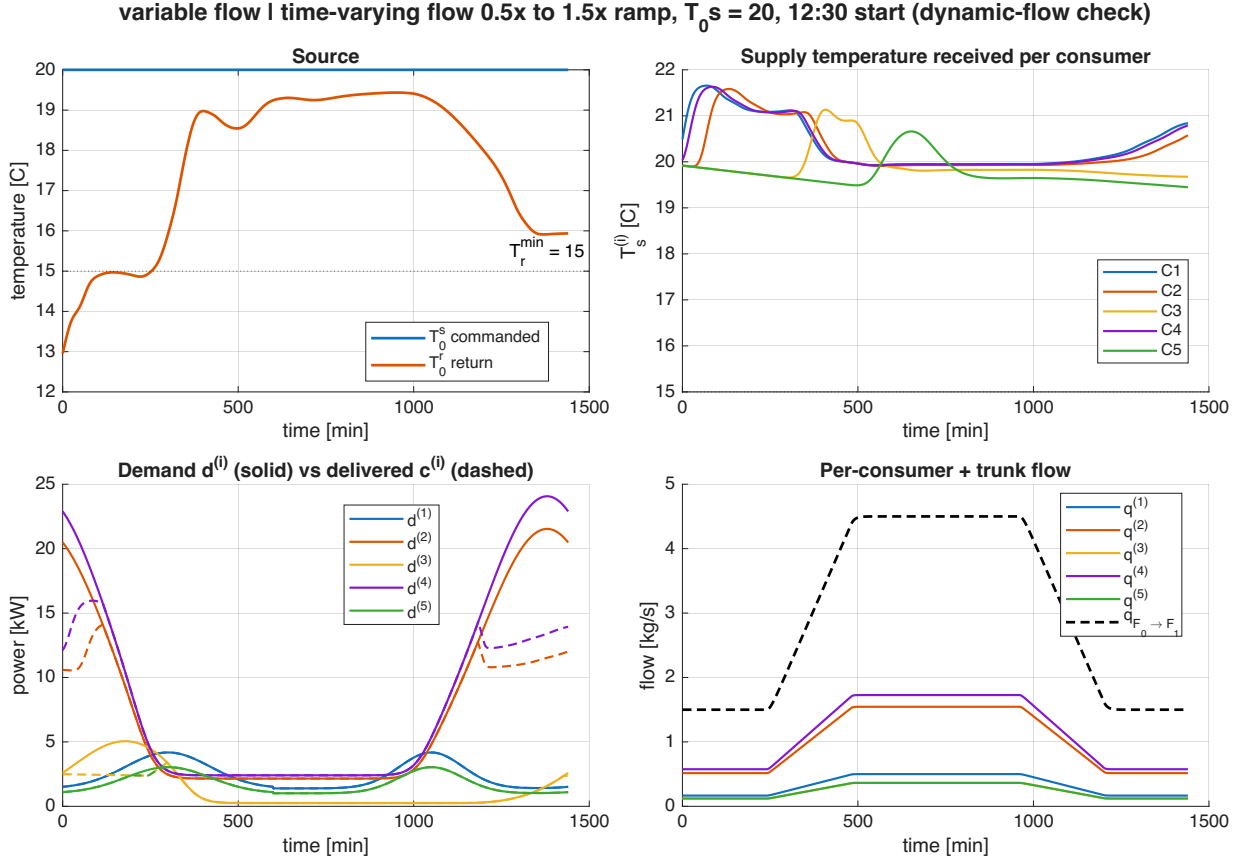


Figure 2: Variable-flow scenario: a commanded $0.5 \rightarrow 1.5 \rightarrow 0.5\times$ ramp; every consumer flow follows its reference smoothly and the trunk carries the sum.

Each consumer flow tracks its ramped reference through the first-order lag. This is $q_{k+1} = aq_k + (1-a)r_{q,k}$ under a ramp, so flows are states, not inputs. The ramp starts and ends at $0.5\times$, where the demand peaks fall, so this scenario meets 81.48 % rather than 100 %.

The four dashboards not shown (`demo_nominal_midday`, `demo_morning_ramp`, `demo_high_flow`, `demo_supply_step`) use the same layout; the supply-step shortfall comes from its 18°C phase cutting $c_{\max}$. `demo_invariants.pdf` collects the invariant margins of all six scenarios (Table 2).

3.3 flow_conservation.pdf: the trunk carries the sum

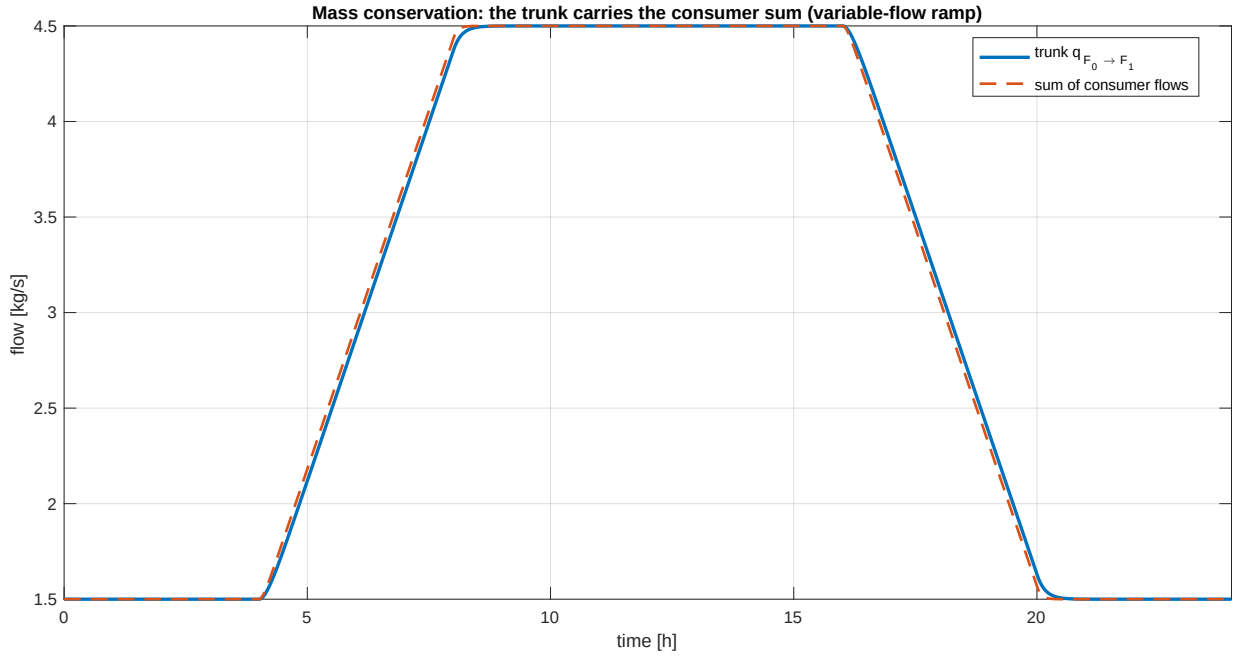


Figure 3: Trunk flow versus the sum of the five consumer flows through the ramp: identical at steady state, briefly apart on the ramps.

At steady state the two curves coincide to machine precision, about 10^{-11} kg/s (the T3/T8 equality). On the ramps they separate by up to 3.9% for a few samples, because each edge lags independently until the lags settle; this is why the conservation invariants are asserted at steady state.

3.4 pulse.pdf: transport delay measured against geometry

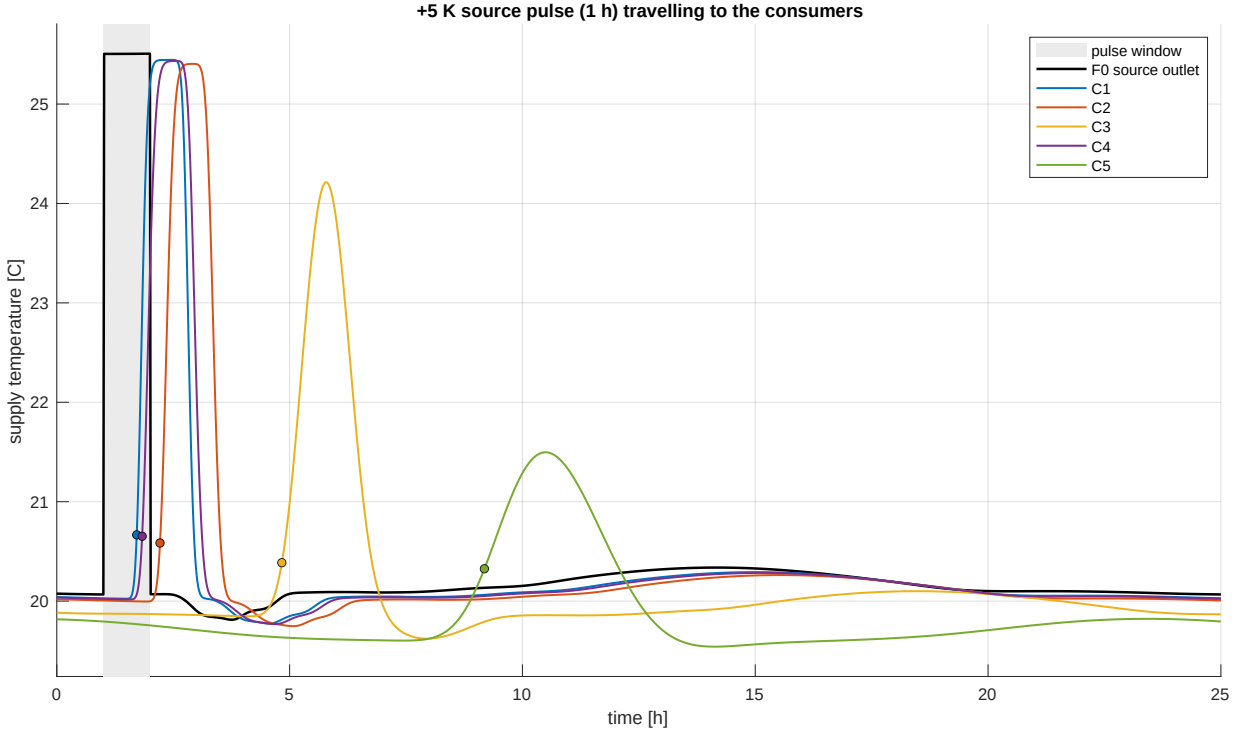


Figure 4: A +5 K one-hour source pulse travels the network: it reaches C1/C4 within the hour and C5 about 8 h later, each peak smaller than the last, and the network settles back to baseline.

The direct test of the advection physics, at constant nominal flows (delays scale $\sim 1/\text{flow}$, about $2.8\times$ longer at the stressed $0.35\times$ point). The expected per-consumer delay is the plug-flow transit time from the geometry,

$$\tau^{(i)} = \sum_{e \in \text{path}(i)} \frac{V_e}{q_e/\rho}, \quad V_e = A_e L_e,$$

pipe volume over volumetric flow along the supply path. Measured arrival must fall within $[0.7, 1.2]$ of that for all five consumers (P1); arrival is detected when a tap first rises 0.5 K (10 % of the pulse). The band is wide on purpose: the 10 m upwind cells smear the front forward, so measured arrivals run ahead of the plug-flow time, and the upper 1.2 leaves room for late detection under junction mixing.

Three things confirm the physics. Ordered arrivals: C5 sits about 8.7 h downstream via its faster feed, 9.8 h via the other, far beyond the ≈ 94 min a naive trunk-velocity estimate gives. Monotone attenuation: the front smears as it advects (C5 is fed 50/50 by two paths about an hour apart, plus a few percent ambient loss). Return to within 0.2 K of baseline at every tap: no artificial energy accumulation. Near consumers barely attenuate (peaks 5.4 K against the 5.44 K realised source pulse) because their transit is shorter than the pulse; the measured peaks fall with distance, which is what a pure delay line without mixing would not do. The transit time scales as $1/q$ through $\tau = \sum_e V_e/(q_e/\rho)$, so raising the flow shortens every delay in the same proportion and leaves the arrival ordering intact.

3.5 delay_parity.pdf: delays match the geometry

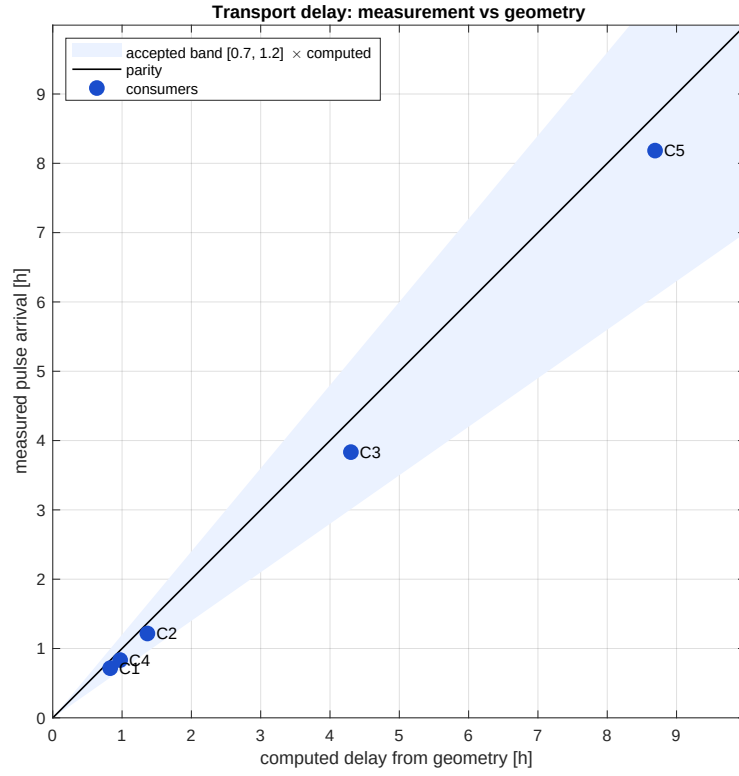


Figure 5: Measured pulse arrival against the delay computed from pipe volume over flow; all five points inside the $[0.7, 1.2]$ band.

The ratios are 0.86 to 0.94, all inside the coded $[0.7, 1.2]$ band, and consistently just below 1 because the detector triggers on the leading edge of the numerically smeared front (Section 3.4).

3.6 verify_ode_resolution.pdf: not a solver artifact

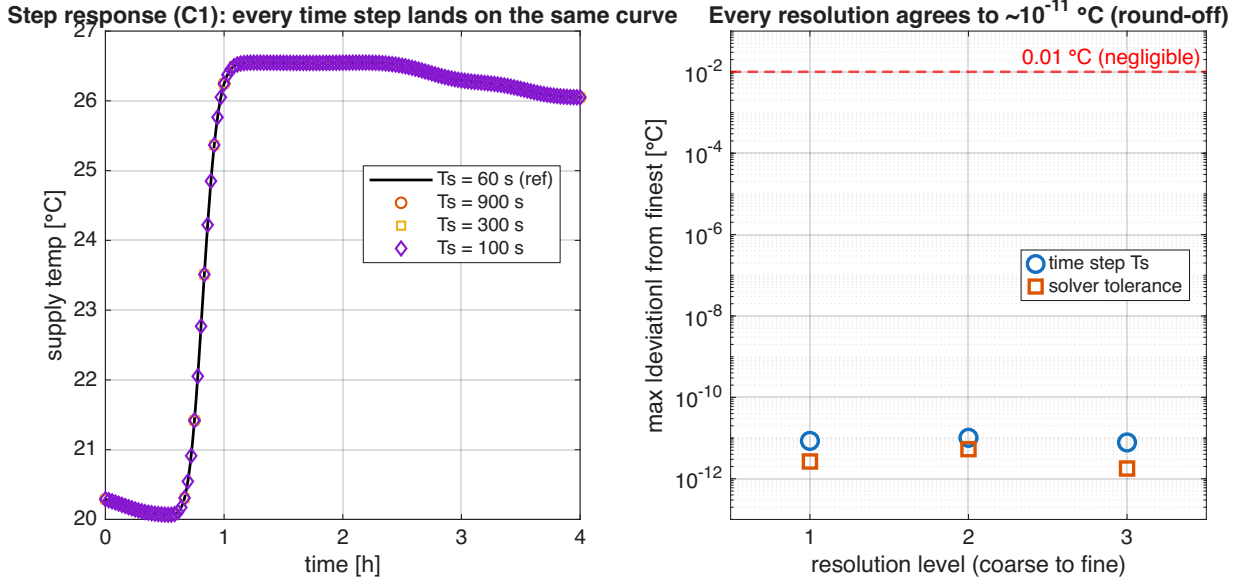


Figure 6: Every time step lands on the same step-response curve; all deviations from the finest setting sit at 10^{-12} to 10^{-11} °C, nine orders below the 0.01 °C negligibility line.

The same +6 °C step is run while refining the sampling step ($T_s = 900 \rightarrow 60$ s, which tightens `ode45 MaxStep`) and, separately, the tolerance (`RelTol` $10^{-4} \rightarrow 10^{-10}$). Consumer temperatures at the common 900s instants move by about 10^{-11} °C, the double-precision floor; both sweeps are hard-asserted below 10^{-6} °C. Right panel: $T_s = 900, 300, 100$ s (circles) and `RelTol` = $10^{-4}, 10^{-6}, 10^{-8}$ (squares) on a fixed 300 s grid. The production defaults ($T_s = 900$ s, `RelTol` = 10^{-6}) are converged, so downstream results stand on physics, not integration error.

The scope of this sweep is worth stating plainly, because it is narrower than the title suggests. It refines the time step and the solver tolerance. It does not refine the 10m spatial cells, so it certifies that the *time* integration is converged, not that the spatial discretisation is. The cell size is a modelling choice, and its cost is the forward front smearing quantified in Section 3.4: measured arrivals run ahead of the geometric transit time, but stay inside the $[0.7, 1.2]$ band the delay-parity check asserts.

4 The automated checks

`tests/validate_plant.m` codes nine invariants and checks all six scenarios of the shipped `demo.mat`. Pass criteria and observed values are kept separate, so documentation and code cannot be played against each other (Table 2). The fresh `run_all_tests` log (MATLAB R2025b, July 2026) shows 9/9 PASS.

Feasibility contract ($T1, T2, T7$). $T1/T2$ confirm the two contract bounds at machine precision. $T7$ is falsifiable positivity: the saved $c^{(i)}$ is zero-clipped, so the raw pre-clip product $c_p(T^{(s)} - T^{(r)})q$ is checked instead, showing the physics never relied on the clip; with $T^{(r)} \leq T^{(s)}$ everywhere, all four bounds hold at every step.

Mass conservation ($T3, T8$). $T8$ applies the full incidence matrices (built independently of the simulation) at every junction, $T3$ the trunk against the consumer sum, both at steady state; the per-edge lags break trunk Kirchhoff by $\approx 3.9\%$ for a few samples after a flow move, documented and deliberately not asserted. A missing `q_edges` field makes $T8$ error, not silently pass.

#	invariant	assert (coded)	observed (<code>demo.mat</code>)
T1	$c^{(i)} \leq d^{(i)}$, every step	violation ≤ 1 W	$\max c - d = 4 \cdot 10^{-11}$ W
T2	$T^{(i,r)} \geq T_{\min}^{(r)} = 15^\circ\text{C}$	violation ≤ 0.1 K	never below the floor (0 K)
T3	trunk Kirchhoff, steady state	$< 10^{-5}$ kg/s	$\leq 6.06 \cdot 10^{-12}$ kg/s
T4	demand-met regime per scenario	per-scenario bands	each scenario inside its band
T5	saved $c = c_p(T^{(s)} - T^{(r)})q$ recomputed	$< 10^{-9}$ W	$3.64 \cdot 10^{-12}$ W
T6	$E_{\text{extracted}}/E_{\text{source}}$	within $[0.5, 1.0]$	$[0.794, 0.965]$
T7	raw pre-clip product; $T^{(r)} \leq T^{(s)}$	> -1 W; violation ≤ 0.1 K	min +253.33 W; max $T^{(r)} - T^{(s)} = -0.167$ K
T8	per-junction Kirchhoff (full incidence)	$< 10^{-9}$ kg/s	$\leq 2.11 \cdot 10^{-12}$ kg/s
T9	source-side return mixing	$< 5\%$ relative	$\leq 2.283 \cdot 10^{-2}$

Table 2: The nine coded invariants: pass criterion versus observed value (worst case over all six scenarios, from `run_all_log.txt` and `figures/demo_invariants.pdf`).

Regime coverage (T4). Capacity-binding scenarios are asserted partial, adequate ones complete, each inside a coded band, so the set spans both sides of the capacity boundary.

Stored data physical (T5, T6, T9). T5 recomputes $c^{(i)}$ from the saved arrays; since the simulator derives $c^{(i)}$ from those same arrays, this is a storage-pipeline consistency check, not energy conservation. T6 puts the extracted-over-source ratio in a range consistent with $\theta_1 = 0.7$ and pipe losses; T9’s tolerance is derived from the loss coefficient, not fitted to the data.

Beyond the nine, `demo_pulse.m` asserts its four transport checks inline, and the resolution script hard-asserts both sweeps below 10^{-6}°C . The flow-dynamics replay in the predictor validator reproduces the saved flows from the coded map with residual 0.00 kg/s, closing the loop between these equations and the code that generated the data.

Taken together: the documented equations are provably the ones that ran, the data respects mass conservation and the substation contract at machine precision, the transport delays match geometry, and none of it depends on the solver settings. That qualifies this plant as the ground truth for everything downstream.